

Mohammad Yehya Hayati

Portfolio: mohammadyehya.github.io
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EDUCATION

- National University of Computer & Emerging Sciences** Karachi, Pakistan
Bachelor of Science - Computer Science; GPA: 3.90/4.00 Aug 2021 - Jun 2025

SKILLS SUMMARY

- Programming Languages:** Python, JavaScript / TypeScript, C/C++, Rust, SQL, Bash, PowerFX
- AI/ML Frameworks & Tools:** Agentic RAG, Multi-Agent Orchestration (LangGraph, LangChain, CrewAI), MCP (Model Context Protocol), Fine-tuning (LoRA/QLoRA), Transformers, Computer Vision, TensorFlow, PyTorch, Scikit-Learn
- Production & MLOps:** LLM Evaluation (RAGAS, DeepEval), Experiment Tracking (MLflow, W&B), Docker, Kubernetes
- Web & Backend Stack:** FastAPI, Next.js, Node.js, Express.js, Socket.IO, TailwindCSS
- Cloud & Infrastructure:** Azure, AWS, GCP, Vector DBs (FAISS, Chroma, Pinecone), Vercel

(Full List on Portfolio)

EXPERIENCE

- Systems Limited** Onsite
Associate Consultant Jul 2025 - Present
Client Engagement: Mashreq Bank

- Chequebook Request Tracking Service:** Engineered an end-to-end chequebook lifecycle management service integrated with the PostEX courier API within a Java/Spring Boot backend, enabling real-time shipment status polling and webhook-driven status reconciliation for 500+ daily chequebook requests, eliminating manual tracking overhead and reducing customer status-inquiry calls by ~25%.
- Multi-Stage Azure DevOps Pipeline Operations:** Operated within a production-grade multi-stage Azure Pipeline infrastructure (Dev → UAT → Prod), managing microservice deployment promotions across controlled environments with pre-configured approval gates and environment-scoped variable groups, developing hands-on fluency in enterprise CI/CD governance and release coordination across cross-functional squads.
- Dual-Gateway OAuth Token Cache Architecture:** Diagnosed and resolved a silent cache-key collision in the Spring Boot token management layer where a shared `ConcurrentHashMap` caused `computeIfAbsent` to return the pre-warmed internal `LoadingCache` for every external OAuth call, resulting in 100% of external API requests being authenticated with internal middleware credentials. Refactored the design by introducing a dedicated `externalTokenCaches` map, cleanly isolating internal and external `LoadingCache` lifecycles, eliminating cross-flow token contamination, and reducing mean token-resolution latency for external calls by removing the redundant cache-miss fallback path, with zero changes to the existing internal auth flow.
- Shared Environment Isolation & Pipeline Migration:** Resolved cross-squad environment contention on a shared Java codebase by migrating the squad's services into a dedicated repository and regenerating Azure Pipelines YAML configurations, enabling fully independent CI/CD execution and eliminating deployment conflicts across teams.

Internal Initiatives

- Agentic AI RFP Automation:** Architected a 6-agent LangGraph RFP system with hierarchical document ingestion across 5+ file types; slashed manual proposal generation from ~40 hrs to under 8 hrs (~80% reduction) while raising technical compliance accuracy by ~30%.
- Enterprise Operational Excellence:** Automated CI/CD pipelines and enterprise testing frameworks across multiple business units, cutting manual deployment overhead by 40%.

- Imperium Dynamics** Onsite
Software Engineer Intern Apr 2025 - Jun 2025

- Pro-Code Extension Development:** Crafted complex C# server-side plugins for the Microsoft Dataverse environment to enforce mission-critical business logic and manage high-volume data transformations with zero data-loss incidents.
- System Integration:** Facilitated seamless data flow across the Microsoft Power Platform by integrating Dynamics 365 with third-party APIs and legacy services cutting manual cross-platform data entry by ~60% and syncing records every 15 minutes.
- Workflow Automation:** Engineered multi-stage Power Automate approval cycles and automated notification systems to streamline administrative workflows.

- Syslab.ai** Onsite
Research Assistant & Full Stack Machine Learning Engineer Nov 2023 - Sep 2024
Syslab Research Lab

- Generate Large Dataset:** Scaled training data from 5,000 to 20,000+ samples via rotation, scaling, and noise augmentation, boosting model robustness by ~15% on held-out test sets.
- Data Vizualization on WebApp:** Designed a Next.js / TailwindCSS analytics dashboard tracking 10+ performance metrics across 4 model variants, cutting per-experiment analysis time by ~50%.

EyeconAI

- Production-Ready WebApp:** Constructed a scalable full-stack SaaS web application using Next.js, Express.js, and FastAPI with 5+ REST endpoints, integrating secure user authentication and cloud-deployed API endpoints.
- Context-Aware Models:** Fine-tuned TensorFlow and PyTorch models on 2,000+ labeled samples using LoRA-style context-aware adaptation, improving real-environment accuracy by ~12% over the baseline.

Toyota Indus Motor Company

- Production Anomaly Detection:** Shipped a real-time defect detection system with YOLOv8 and PyTorch achieving 98.7% accuracy at 30 FPS across 4-9 simultaneous camera feeds, surfaced on a Next.js dashboard via a FastAPI streaming backend.

RESEARCH PUBLICATIONS

- **Context-Aware Detection of Mixed Critical Events using Video Classification:** Published on arxiv - arxiv.org/abs/2411.15773v1
 - Developed a novel context-aware system for smart city safety, achieving 99.25% F1-score on traffic incident detection and 91.59% F1-score on fire detection.
 - Utilized DenseNet-Transformer architectures to process spatial-temporal features, outperforming standard CNN baselines by 3% in prediction accuracy.

CERTIFICATIONS

- **Microsoft Azure Fundamentals:** Gained foundational knowledge of cloud concepts, core Azure services, plus Azure management and governance features and tools.  Verify
- **Microsoft Azure Data Fundamentals:** Gained foundational knowledge of core data concepts related to Microsoft Azure data services.  Verify
- **Microsoft Azure AI Fundamentals:** Gained fundamental AI concepts related to the development of software and services of Microsoft Azure to create AI solutions.  Verify
- **Microsoft AI Business Professional:** Gained foundational knowledge of how artificial intelligence can be applied in business scenarios, including AI workloads, responsible AI principles, and identifying opportunities for AI-driven solutions.  Verify
- **Microsoft AI Transformation Leader:** Gained knowledge of how organizations can lead AI-driven transformation, including strategies for responsible AI adoption, change management, and scaling AI initiatives across business units.  Verify
- **GitHub Foundations:** Gained foundational knowledge of GitHub concepts including repository management, collaboration workflows, version control with Git, and core features of the GitHub platform.  Verify
- **GitHub Actions:** Gained knowledge of automating software workflows using GitHub Actions, including building CI/CD pipelines, managing workflow triggers, runners, secrets, and reusable actions.  Verify

PROJECTS

- **VoltOS:** Arch Linux-based distro prioritizing speed and customization, with Quickshell-driven fluid animations.
 - **Agentic OS Interaction:** Engineered a unified AI desktop environment featuring a built-in chatbot with 12+ Model Context Protocol (MCP) tool-bindings, enabling the LLM to execute system-level tool-calling and device management.
 - **Multimodal AI Integration:** Implemented real-time SAM (Segment Anything Model) features for image segment layering and completion, alongside Speech-to-Text (STT) typing and Image Super Resolution.
 - **System Control Architecture:** Shipped a multi-module Quickshell suite which includes, Network Manager (VPN/Firewall), Media Control, cron-style Workflow Builder, live System Stats, and Clipboard Manager, replacing multiple third-party utilities and reducing idle memory by ~20 MB.
 - **Integrated Services:** Feature-rich environment with LAN chat/sharing, Obs-based Recording Center, and dynamic Alt+Tab task switching consolidating 6+ daily workflows into a single cohesive environment.
- **LinguaLink:** Multilingual chat platform with real-time AI translation, built on Next.js, Express.js, FastAPI, and Socket.io.
 - **Real-time Synchronization:** Sustained <800ms average English-to-French translation latency under 50-concurrent-user load testing via a bidirectional Socket.io connection.
 - **Model Optimization:** Trained custom Seq2Seq and Transformer models on 500,000+ parallel sentence pairs, reaching a 81%+ BLEU score in resource-constrained inference.
- **CoinScope:** Real-time cryptocurrency tracker using Kafka for data streaming and Socket.io for live dashboard updates.
 - **Event-Driven Architecture:** Decoupled market data producers from frontend consumers with an Apache Kafka pipeline sustaining 10,000+ price events/second across 50+ tracked assets with <500ms end-to-end dashboard refresh latency.

HONORS AND AWARDS

- Graduated Summa Cum Laude (Top 0.5%) - Jun 2025
- Placed in Rector's List (4 SGPA) x 3 - Jan 2022, Jan 2023, May 2023
- Placed in Dean's List (3.5+ SGPA) x 5 - May 2022, Jan 2024, May 2024, Jan 2025, May 2025
- First Place at Database Odyssey Competition - Mar 2025
- First Place at Openhouse Database Project Competition - Dec 2024
- First Place at Procom Math Olympiad Competition - Apr 2023
- Runner's Up at Coder's Cup Speed Programming Competition - Nov 2021
- Achieved a score of 8 in Academic IELTS - Nov 2025